

# A Two-Stage Deep Learning Cascade Framework for Robust Papaya Leaf Disease Detection and Classification

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**Abstract**— Papaya leaf diseases can significantly reduce crop yield and quality, while manual disease diagnosis is often time-consuming, subjective, and affected by complex field conditions. This study proposes a two-stage deep learning cascade framework for papaya leaf disease detection and classification. In the first stage, YOLOv11m is used to localize suspicious disease regions and extract Regions of Interest from leaf images. In the second stage, EfficientNet-B2 performs ROI-based classification into five categories: Healthy, Ring Spot, Curl, Bacterial Spot, and Anthracnose. The proposed framework was evaluated using the BDPapayaLeaf dataset, with separate experiments conducted for disease-region detection and final five-class classification. To reduce the effect of class imbalance, class-weighted loss, oversampling, and data augmentation were applied during the classification stage. Experimental results show that the YOLOv11m detector achieved an overall mAP@0.5 of 0.777 on the four disease classes. For the complete cascade framework, the proposed YOLOv11m–EfficientNet-B2 model achieved 94.85% accuracy, 94.92% macro precision, 94.93% macro recall, and 94.92% macro F1-score on the final test set of 291 images. Compared with the single-stage EfficientNet-B2 baseline, the cascade framework improved classification performance by reducing background interference and allowing the classifier to focus on localized disease symptoms. These results indicate that the proposed framework is a practical and effective approach for supporting automated papaya leaf disease diagnosis.

**Keywords:** Deep Learning, Papaya Leaf Disease Detection, Smart Agriculture

## I. INTRODUCTION

Papaya is a nutritious fruit that contains many important nutrients such as fiber, antioxidants, and vitamin C. It is beneficial for human health and is widely grown in many agricultural regions [1]. However, papaya plants are vulnerable to many diseases that can reduce both yield and fruit quality [2]. Common diseases such as anthracnose, bacterial spot, leaf curl, and ring spot can damage the leaves

and later affect the fruits. This causes economic losses for farmers and reduces the market value of the crop.

One major problem is that the early symptoms of papaya leaf diseases often look similar to nutrient deficiency symptoms. Because of this, farmers may find it difficult to identify the disease correctly by visual observation alone. In many cases, they still rely on manual inspection and spray pesticides [3]. This process is slow, subjective, and sometimes inaccurate. Excessive or incorrect use of pesticides can also harm the crops, pollute soil and water, and negatively affect human health and the environment.

To address these issues, early and accurate disease detection is very important. In recent years, computer vision and deep learning have become useful tools for automatic plant disease detection [4]–[5]. In this study, we propose a papaya leaf disease detection system based on YOLO and CNN. First, YOLO is used to detect diseased regions on papaya leaves. Then, a CNN model is used to classify the detected disease into specific categories. This approach can help improve detection efficiency and support a faster, more practical solution for disease diagnosis in papaya farming.

## II. THEORETICAL BACKGROUND AND RELATED WORKS

### A. Convolutional Neural Networks

CNNs are widely used in computer vision because they can learn visual features directly from raw image pixels instead of relying on manually designed features. This makes them suitable for agricultural image analysis, especially plant leaf disease classification [6].

A typical CNN includes convolutional layers, pooling layers, and classification layers. Convolutional layers extract local patterns such as edges, textures, shapes, and color variations [6]. Pooling layers reduce feature map size and computational cost while retaining important information. The final layers use these learned features to classify the image into target categories. With local connectivity, shared weights,

downsampling, and nonlinear activation functions such as ReLU, CNNs can learn complex visual differences and serve as a strong baseline for detecting subtle disease symptoms on plant leaves.

### B. EfficientNet

EfficientNet [20] introduces a principled compound scaling method that jointly scales network depth, width, and input resolution, enabling improved accuracy–efficiency trade-offs. Based on this approach, a family of models (B0–B7) is constructed, where larger variants achieve higher performance with significantly better parameter and FLOPS efficiency compared to conventional CNN architectures.

The main idea of EfficientNet is compound scaling, which scales network depth, width, and input resolution in a balanced way [7]. Instead of increasing only one dimension, EfficientNet uses a compound coefficient to control all three factors together. This design allows the model to achieve strong performance while remaining efficient, making it suitable for plant leaf disease classification tasks:

$$\begin{aligned} \text{depth: } d &= \alpha^\varphi \\ \text{width: } w &= \beta^\varphi \\ \text{resolution: } r &= \gamma^\varphi \end{aligned}$$

where  $\alpha$ ,  $\beta$ , and  $\gamma$  are constants determined by grid search, and  $\varphi$  is a user-defined coefficient. Here,  $\alpha$  controls the scaling of network depth,  $\beta$  controls the scaling of network width, and  $\gamma$  controls the scaling of input image resolution. These coefficients are usually constrained such that  $\alpha \geq 1$ ,  $\beta \geq 1$ , and  $\gamma \geq 1$ , and  $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$ , ensuring a balanced increase in computational cost.

Another important component of EfficientNet is the use of Mobile Inverted Bottleneck Convolution (MBConv) blocks, which were originally introduced in MobileNetV2. These blocks first expand the number of channels and then project them back to a lower-dimensional space, enabling efficient feature extraction. Additionally, MBConv uses depthwise separable convolution, which reduces computational cost by approximately  $k^2$  times compared to standard convolution, where  $k$  is the kernel size.

### C. YOLO

YOLO, which stands for You Only Look Once, is a deep learning-based object detection framework designed to identify and localize objects in an image using a single forward pass [8]–[9]. The main idea of YOLO is to formulate object detection as a regression problem [8], [10]. Instead of scanning an image multiple times or analyzing several proposed regions separately, YOLO directly predicts bounding box coordinates, objectness scores, and class probabilities from the entire input image. The image is divided into a grid, and each grid cell is responsible for detecting objects whose centers fall inside that cell [9]. This design enables YOLO to use global image information, which helps the model understand the overall context of the scene.

In a typical YOLO architecture, the network consists of three main components: the backbone, the neck, and the head [8]. The backbone is responsible for extracting visual features from the input image through a series of convolutional and pooling layers. As shown in the architecture diagram, the input image is gradually downsampled while the number of feature channels increases. This allows the network to learn both low-level features, such as edges and textures, and higher-level semantic features related to object regions.

### D. Related Works

Several studies have demonstrated the effectiveness of image processing, computer vision, and deep learning techniques in agricultural recognition and classification tasks. Recent research has increasingly adopted deep learning models for practical agricultural applications. A study in [11] investigated the recognition of 41 medicinal plants using CNN-based models on mobile platforms, where EfficientNet-B0 combined with transfer learning and data augmentation achieved an accuracy of approximately 94%. In [12], an automated fruit-sorting system based on YOLOv3 was proposed for tomato quality classification on conveyor belts, achieving about 94% accuracy. Similarly, [13] applied an improved YOLOv7 model for tomato quality assessment, obtaining 93.3% accuracy for normal tomatoes and 89.1% for defective ones.

Deep learning has also been widely applied to papaya disease detection and plant health diagnosis. A transfer learning-based CNN approach for papaya leaf disease recognition was introduced in [14]. In [15], a benchmark dataset for papaya leaf disease detection and classification was developed, enabling the evaluation of modern architectures such as YOLOv8 and Vision Transformers. A comparative study in [16] evaluated multiple transfer learning models, including VGG-16, ResNet-50, DenseNet-121, and InceptionV4, demonstrating that pretrained CNNs combined with data augmentation can significantly improve classification performance while mitigating overfitting. In [17], a deep CNN-based framework for fruit and vegetable recognition under complex real-world conditions achieved an accuracy of 96% across 20 categories. Furthermore, a YOLO-based approach for papaya disease detection was proposed in [18], highlighting its applicability to large-scale detection tasks, including post-harvest quality control, embedded systems, and smart agriculture.

Overall, existing studies confirm that CNN- and YOLO-based approaches are effective for agricultural image recognition and plant disease detection. However, most prior works primarily focus on either whole-image classification or single-stage object detection, which remain susceptible to challenges such as complex backgrounds, illumination variations, and inter-class visual similarity among disease symptoms. To address these limitations, this study proposes a two-stage cascade framework that integrates YOLOv11m [19] for disease region localization with EfficientNet-B2 [20] for region-based classification, aiming to improve the accuracy and robustness of papaya leaf disease recognition.

## III. METHODOLOGY

### A. System Architecture

This study evaluates a two-stage deep learning cascade framework that combines YOLOv11m and EfficientNet-B2 for papaya leaf disease detection and classification. YOLOv11m is used to localize suspicious disease regions, while EfficientNet-B2 classifies the extracted ROIs into specific disease categories.

The main purpose of using a two-stage framework is to reduce the influence of complex backgrounds. In real-world leaf images, the background may include soil, branches, other leaves, or lighting variations. If the whole image is directly passed into a classifier, these irrelevant regions may reduce classification performance. Therefore, YOLOv11 first extracts the important Regions of Interest (ROIs), and

EfficientNet-B2 then focuses only on these regions for classification. The proposed system classifies papaya leaf images into five categories: Healthy, Ring Spot, Curl, Bacterial Spot, and Anthracnose. The overall pipeline of the proposed method is shown in Fig. 1.

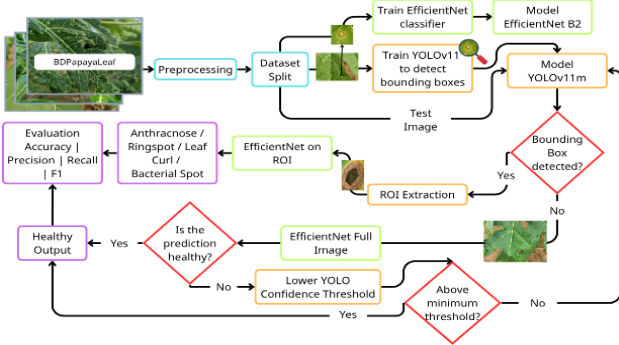


Fig. 1. The overall pipeline of the proposed two-stage framework.

### B. Dataset description, preprocessing, and augmentation

In the first phase, the BDPapayaLeaf dataset was prepared for training and evaluation. The dataset contains papaya leaf images belonging to four disease classes and one healthy class. The disease classes include Ring Spot, Curl, Bacterial Spot, and Anthracnose, while the healthy class contains images of normal papaya leaves.

The dataset was divided into three subsets: training, validation, and testing. The training set was used to train the models, the validation set was used to monitor model performance during training, and the test set was used for final evaluation.

In this study, the dataset was split using a 70:15:15 ratio, where 70% of the data was used for training, 15% for validation, and 15% for testing. This split was applied to both the YOLOv11 detection dataset and the EfficientNet-B2 classification dataset to ensure a consistent evaluation process.

TABLE I. DISTRIBUTION OF THE YOLOV11 DETECTION DATASET.

| Class          | Total Images | Train | Validation | Test |
|----------------|--------------|-------|------------|------|
| Anthracnose    | 355          | 248   | 53         | 54   |
| Bacterial Spot | 458          | 320   | 69         | 69   |
| Curl           | 361          | 253   | 54         | 54   |
| Ring Spot      | 533          | 373   | 80         | 80   |
| Total          | 1707         | 1194  | 256        | 257  |

For the YOLOv11 detection stage, the disease images were prepared in YOLO format, including image files and corresponding bounding box annotation files. Each annotation file contains the class label and the normalized coordinates of the diseased region. The healthy images were not assigned bounding boxes because they do not contain visible disease regions.

TABLE II. DISTRIBUTION OF THE EFFICIENTNET-B2 CLASSIFICATION DATASET.

| Class          | Total Images | Train | Validation | Test |
|----------------|--------------|-------|------------|------|
| Anthracnose    | 6502         | 4416  | 1046       | 1040 |
| Bacterial Spot | 458          | 320   | 69         | 69   |
| Curl           | 362          | 254   | 54         | 54   |
| Healthy        | 228          | 160   | 34         | 34   |
| Ring Spot      | 3561         | 2412  | 510        | 639  |
| Total          | 11111        | 7562  | 1713       | 1836 |

For the EfficientNet-B2 classification stage, the images were prepared as class-based folders. The diseased regions detected by YOLOv11 were cropped and used as input images for the CNN classifier. Health images were used as full images because they do not contain bounding box annotations.

After constructing the EfficientNet-B2 classification dataset, class imbalance was addressed using class-weighted sparse categorical cross-entropy. The class weight for class  $c$  was calculated as  $w_c = \frac{N}{Kn_c}$  where  $N$  is the total number of training samples,  $K$  is the number of classes, and  $n_c$  is the number of training samples belonging to class  $c$ . Based on the training distribution shown in Table II, the resulting weights were 0.3425 for Anthracnose, 4.7263 for Bacterial Spot, 5.9543 for Curl, 9.4525 for Healthy, and 0.6270 for Ring Spot. Therefore, errors involving minority classes, particularly Healthy, Curl, and Bacterial Spot, contributed more strongly to the training loss than errors involving majority classes. Oversampling and data augmentation were also applied during training.

The final evaluation was conducted by using a combined test set for the complete YOLOv11–EfficientNet-B2 cascade framework. Specifically, 257 test images from the YOLO dataset were used for the four disease classes, while 34 additional Healthy images were included for CNN-based classification because healthy leaves do not contain bounding-box annotations. Therefore, the final test set consisted of 291 images and represented the five-class classification task rather than the YOLO detection dataset alone.

### C. Detection phase

In this study, YOLOv11m is selected as the object detection model due to its superior balance between accuracy, efficiency, and real-time performance compared to earlier object detection approaches [9]. Among the YOLO family, YOLOv11 represents the latest advancement, incorporating several architectural improvements over previous versions. One key reason for choosing YOLOv11 is its enhanced feature extraction capability, achieved through newly introduced modules such as the C3k2 block and C2PSA attention mechanism. These components allow the model to capture more discriminative spatial features and focus on important regions in the image, which is particularly beneficial for detecting small or subtle objects such as disease spots on leaves.

In addition, YOLOv11 demonstrates better multi-scale detection performance, which is critical in agricultural datasets where objects (e.g., disease regions) vary in size and shape [19]. Its architecture effectively combines features at different resolutions, improving detection robustness across varying conditions.

### D. ROI Extraction

After YOLOv11 predicts the bounding boxes, the corresponding disease regions are cropped from the original image. Each cropped region is considered a Region of Interest (ROI). This step is important because it removes unnecessary background information and allows the classification model to focus on the disease symptoms.

The extracted ROIs are resized to match the input size required by EfficientNet-B2. If multiple bounding boxes are detected in one image, each region can be classified separately. The final prediction can then be selected based on

the highest confidence score or the most relevant detected region.

#### E. Classification phase

EfficientNet-B2 was selected as the classification model in this study due to its favorable trade-off between accuracy, model complexity, and computational cost [20]. In the proposed YOLO-CNN pipeline, YOLOv11 is employed to localize disease regions on papaya leaves, after which EfficientNet-B2 performs classification on the extracted regions of interest (ROIs). Therefore, the classifier must exhibit strong feature representation capability to capture subtle disease patterns while maintaining computational efficiency.

The suitability of EfficientNet-B2 for meeting these lightweight operational requirements is clearly demonstrated by its baseline performance metrics. As reported in [20], EfficientNet-B2 achieves a Top-1 accuracy of 80.1% on the ImageNet dataset with only 9.2 million parameters and 1.0 billion FLOPs. When compared to traditional Convolutional Neural Networks with similar accuracy levels, the efficiency of EfficientNet-B2 becomes highly evident. For instance, models like Inception-v4 (80.0% accuracy) and Inception-resnet-v2 (80.1% accuracy) require between 48 and 56 million parameters and 13 billion FLOPs. This means EfficientNet-B2 achieves the same predictive performance while reducing the parameter size by a factor of 5 to 6, and decreasing the computational cost (FLOPs) by 13 times.

This significant reduction in computational cost without sacrificing accuracy is achieved through the model's unique structural design. The architecture of EfficientNet-B2 relies on Mobile Inverted Residual Bottleneck blocks, which are efficient convolutional structures that support complex visual feature learning while reducing computational demands.

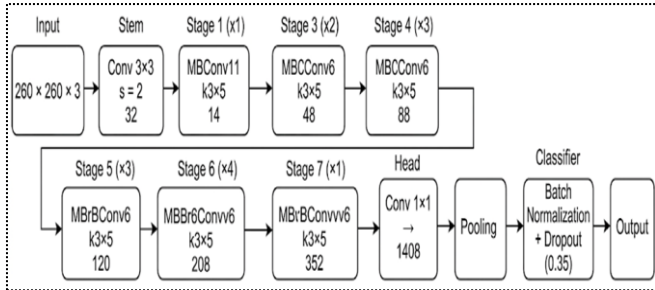


Fig. 2. EfficientNet B2 Architecture

#### F. Evaluation Metrics

The performance of the proposed framework was evaluated using detection and classification metrics. For the YOLOv11m detection stage, Precision, Recall, mAP@0.5, and mAP@0.5:0.95 were used to measure the localization performance of disease regions. For the EfficientNet-B2 classification stage and the complete cascade framework, Accuracy, Precision, Recall, F1-score, and the confusion matrix were used to evaluate five-class classification performance.

Accuracy measures the overall proportion of correctly classified samples, while Precision and Recall evaluate the model's false positive and false negative behavior. F1-score is the harmonic mean of Precision and Recall and provides a balanced evaluation, especially for imbalanced classes. The main classification metrics are defined as follows:

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1\text{-score} = 2 * \frac{Precision * Recall}{Precision + Recall}$$

where TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

#### IV. EXPERIMENTS AND RESULTS

Before evaluating the complete cascade framework, the YOLOv11m detector was first tested to assess its ability to localize disease regions on papaya leaves. Since the Healthy class does not contain disease regions, it was not included in the detection evaluation. Therefore, the YOLOv11m test set consisted of four disease classes only.

As shown in Table III, YOLOv11m achieved good overall detection performance on the test set. The model performed particularly well on Bacterial Spot and Curl, while lower results were observed for Anthracnose and Ring Spot. This difference may be due to variations in lesion size, irregular disease distribution, and visual similarity between disease symptoms. Overall, the detection results indicate that YOLOv11m can provide useful localized regions for the second-stage EfficientNet-B2 classifier.

TABLE III. DETECTION PERFORMANCE OF YOLOV11M ON THE TEST SET

| Class          | Precision | Recall | mAP@0.5 | mAP@0.5:0.95 |
|----------------|-----------|--------|---------|--------------|
| Anthracnose    | 0.578     | 0.621  | 0.515   | 0.319        |
| Bacterial Spot | 0.958     | 1.000  | 0.993   | 0.915        |
| Curl           | 0.982     | 1.000  | 0.995   | 0.948        |
| Ring Spot      | 0.692     | 0.643  | 0.604   | 0.426        |
| All classes    | 0.802     | 0.816  | 0.777   | 0.652        |

The proposed YOLOv11-EfficientNet-B2 cascade model was evaluated on a test set of 291 images, achieving 276 correct predictions and 15 misclassifications, corresponding to an overall accuracy of approximately 94.85%. This result demonstrates the effectiveness of the proposed two-stage framework in improving classification performance by combining object detection and region-based classification.

In terms of class-wise performance, the model achieved high accuracy across all categories, including Anthracnose (94.44%), Bacterial Spot (94.20%), Curl (98.15%), Healthy (94.12%), and Ring Spot (93.75%). Among these, the Curl class achieved the highest accuracy, indicating that its visual features are more distinctive.

The confusion matrix of the proposed YOLOv11m-EfficientNet-B2 model, presented in Fig. 3, shows that most samples were correctly classified along the diagonal, with only a few misclassifications among visually similar classes. This concentrated diagonal distribution indicates strong classification stability on the tested samples.

To examine the effect of disease-region localization, a single-stage EfficientNet-B2 baseline was first evaluated using the original whole images as input. This baseline does not use YOLOv11m for ROI extraction; therefore, it is reported separately from the cascade models. As shown in Table IV, the EfficientNet-B2 whole-image classifier achieved an accuracy of 92.78%, macro precision of 93.17%,

macro recall of 92.85%, and macro F1-score of 92.93%. Although this result is relatively strong, whole-image classification may still be affected by complex agricultural backgrounds, such as soil, overlapping leaves, branches, and lighting variations.

TABLE IV. PERFORMANCE OF THE SINGLE-STAGE EFFICIENTNET-B2 BASELINE

| Algorithm       | Accuracy (%) | Precision (Macro, %) | Recall (Macro, %) | F1-Score (Macro, (%)) |
|-----------------|--------------|----------------------|-------------------|-----------------------|
| EfficientNet B2 | 92.78        | 93.17                | 92.85             | 92.93                 |

Table V compares different CNN backbones within the same YOLOv11m-based cascade framework. In this setting, YOLOv11m is consistently used as the first-stage detector to extract Regions of Interest, while the second-stage CNN classifier is changed to evaluate the effect of different backbone architectures. This makes the comparison more consistent because all models in Table V follow the same two-stage detection-classification pipeline.

Among the evaluated cascade models, YOLOv11m combined with VGG16 achieved an accuracy of 93.47%. DenseNet121 and ResNet50 produced stronger results, reaching accuracies of 94.16% and 94.50%, respectively. The proposed YOLOv11m–EfficientNet-B2 model achieved the best overall performance, with an accuracy of 94.85%, macro precision of 94.92%, macro recall of 94.93%, and macro F1-score of 94.92%. These results suggest that EfficientNet-B2 is an effective second-stage classifier for localized papaya leaf disease regions.

TABLE V. COMPARISON OF CNN BACKBONES WITHIN THE YOLOV11M-BASED CASCADE FRAMEWORK

| Algorithm             | Accuracy (%) | Precision (Macro, %) | Recall (Macro, %) | F1-Score (Macro, (%)) |
|-----------------------|--------------|----------------------|-------------------|-----------------------|
| YOLOv11m +DenseNet121 | 94.16        | 93.72                | 94.73             | 94.09                 |
| YOLOv11m +ResNet50    | 94.50        | 93.93                | 94.76             | 94.26                 |
| YOLOv11m +VGG16       | 93.47        | 92.88                | 94.23             | 93.28                 |
| <b>Proposed model</b> | <b>94.85</b> | <b>0.9492</b>        | <b>94.93</b>      | <b>94.92</b>          |

Compared with the single-stage EfficientNet-B2 baseline in Table IV, the YOLOv11m–EfficientNet-B2 cascade achieved higher classification performance. This improvement indicates that detecting and cropping disease regions before classification can reduce background interference and help the classifier focus more effectively on visual disease symptoms. Therefore, the proposed framework does not only compare different CNN backbones, but also demonstrates the practical benefit of combining disease-region localization with ROI-based classification.

To further contextualize the obtained results, the proposed model was compared with several existing studies reported in the literature [21]. It should be noted that these studies may differ in terms of dataset size, disease categories, image acquisition conditions, and evaluation protocols; therefore, the comparison should be interpreted as a reference rather than a strictly identical experimental comparison. In a transfer

learning study that evaluated models such as VGG16, DenseNet121, and EfficientNet-B0, the highest reported accuracy was about 87.95%. The proposed model achieved higher reported accuracy, which suggests that combining YOLO-based region detection with CNN classification can be beneficial compared with whole-image CNN classification. The proposed method also achieved slightly higher reported accuracy, precision, recall, and F1-score compared with the HASPNet approach, which reported a validation accuracy of 93.87% [22]. Compared with the Darknet-based study, where the F1-scores varied among classes, such as 0.721 for Bacterial Spot and 0.765 for Curl, the proposed model showed more balanced performance across disease categories in the present experiment [23]. Table VI presents a comparison between the proposed model and existing models reported in previous studies, indicating that the proposed framework achieves competitive performance and shows potential for robust papaya leaf disease classification.

TABLE VI. COMPARISON WITH EXISTING STUDIES

| Model                 | Accuracy (%) | Precision (%) | Recall (%)   | F1-score (%) |
|-----------------------|--------------|---------------|--------------|--------------|
| ResNet50 [21]         | 87.95        | 90.16         | 90.70        | 90.17        |
| HASPNet [22]          | 93.87        | 94.00         | 94.00        | 94.00        |
| Darknet53 [23]        | 84.78        | 85.86         | 86.90        | 86.00        |
| <b>Proposed model</b> | <b>94.85</b> | <b>94.92</b>  | <b>94.93</b> | <b>94.92</b> |

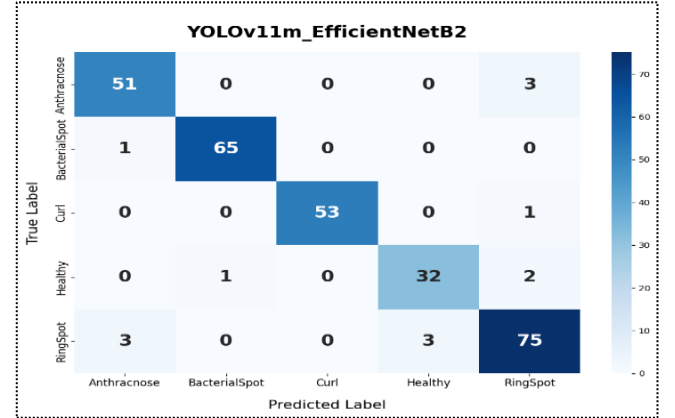


Fig. 3. Confusion matrix of the proposed model.

Overall, these results suggest that the proposed cascade framework is a promising and effective approach for papaya leaf disease classification, particularly when disease localization is used to support region-based classification.

## V. CONCLUSION

This study proposed a two-stage cascade deep learning framework for papaya leaf disease detection and classification by combining YOLOv11 and EfficientNet-B2. The main objective was to improve classification accuracy by first localizing disease regions and then performing region-based classification.

Experimental results demonstrated that the proposed method achieved high performance, with an overall accuracy of approximately 94.85% on the test set. The model showed consistent results across all disease classes and effectively reduced background noise by focusing on Regions of Interest (ROIs). Compared with the single-stage EfficientNet-B2 baseline under the same experimental setting, the proposed

cascade achieved better classification performance and showed competitive results compared with previously reported papaya leaf disease studies.

However, some challenges still remain. Certain disease classes have similar visual symptoms, such as color changes, spots, or irregular leaf patterns, which may lead to misclassification. In addition, image quality, lighting conditions, complex backgrounds, and different leaf orientations can affect the model's performance.

In future work, the proposed system can be improved by using a larger and more diverse dataset collected from different environments. Advanced attention mechanisms may also be integrated to help the model focus more effectively on important disease features. Moreover, more robust architectures or ensemble learning methods can be explored to further improve classification accuracy and generalization performance.

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